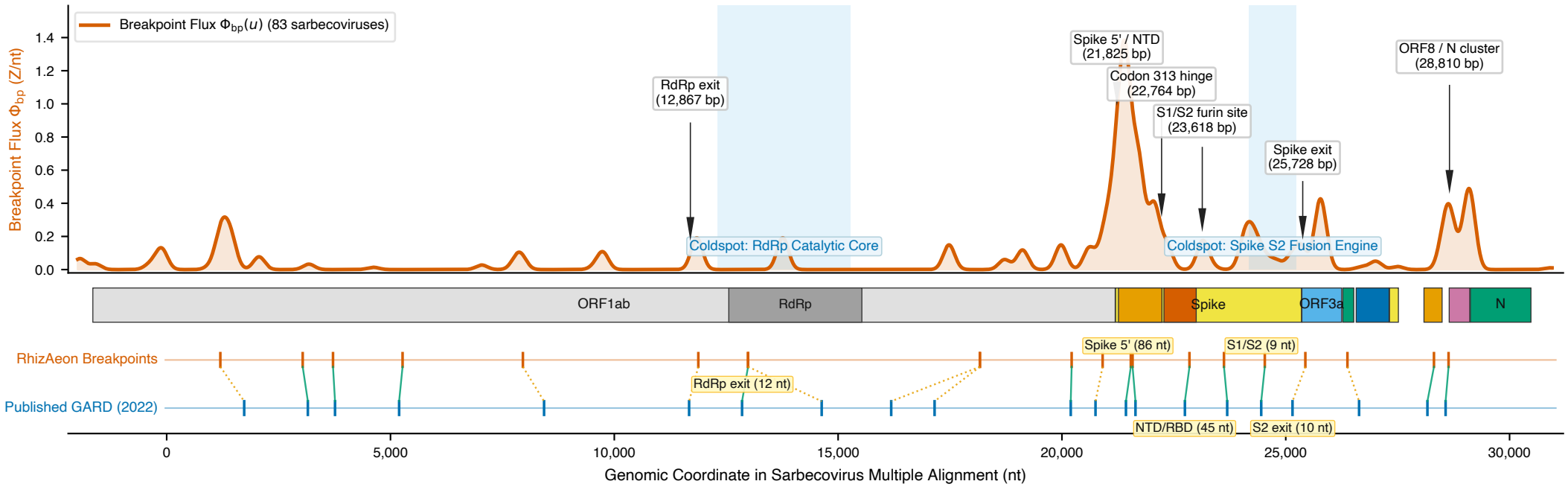
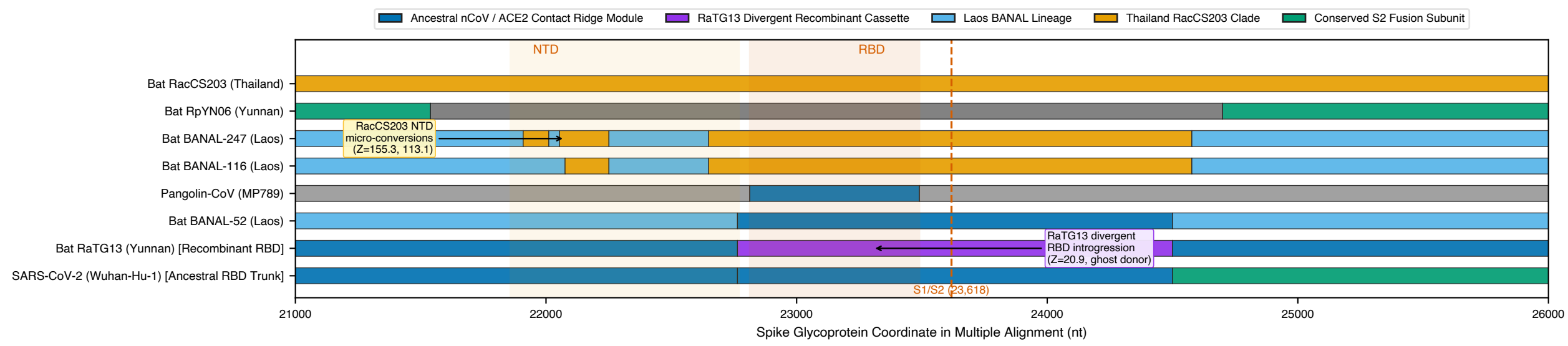
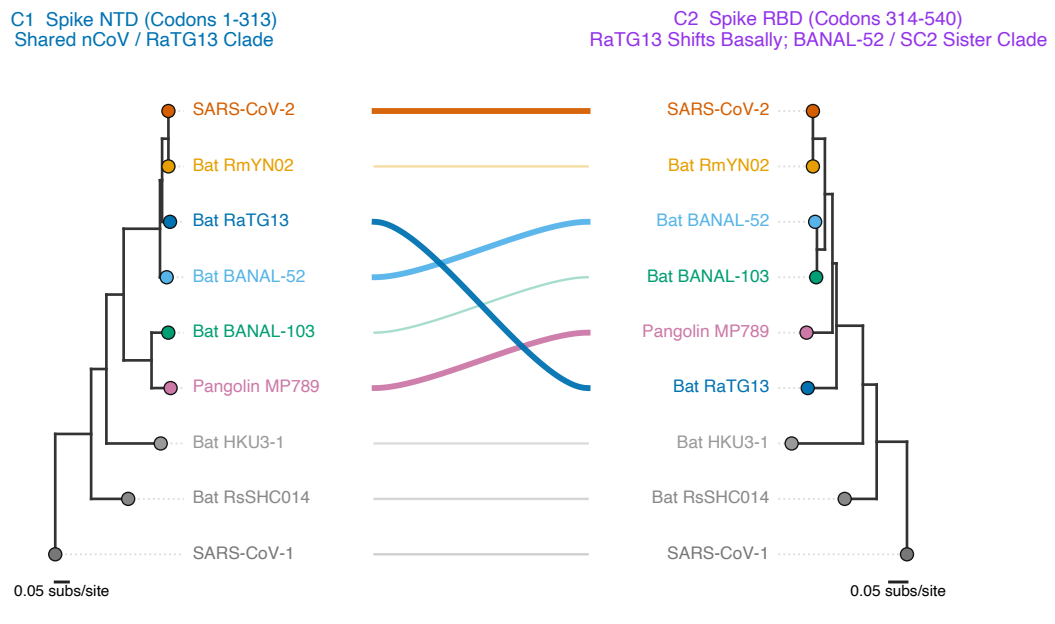
A Genome-wide Recombination Flux $\Phi_{bp}(u)$ and Benchmark Concordance Across 30.9 kb




B Spike Reticulate Mosaic Architecture Across Wildlife Progenitors and SARS-CoV-2




C SARS-CoV-2 / RaTG13 Incongruence Across Codon 313 Hinge




D Polarized Ancestral Recombination Graph (ARG) for Sarbecovirus Progenitors

